

Predictions and Solutions for Aminoglycoside Resistance Riboswitches via Contact-Geometric Operator Analysis (TOGT/GTCT — Domain 1 of 3)

Principia Orthogona — Translational Applications

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Abstract

Aminoglycoside-sensing riboswitches in the 5' leader of *aac/aad* resistance genes [a, i] undergo an inducible conformational switch — OFF (SD₂ sequestered) to ON (SD₂ exposed) — that activates resistance enzyme expression only in the presence of drug. No crystal structure of the full ON state exists; the switching kinetics have not been measured directly; and the universality of the switch across integron contexts is not established.

We apply the TOGT/GTCT contact-geometric operator framework [o, r] to map the riboswitch conformational cycle to the operator chain $\mathbf{G} = \mathbf{U} \circ \mathbf{F} \circ \mathbf{K} \circ \mathbf{C}$ on a dm^3 contact manifold. The fold operator $\mathbf{F}$ (Whitney A_1 singularity) maps to the stem competition bifurcation; the curvature threshold κ^* maps to the free-energy barrier between OFF and ON stems.

We derive three falsifiable quantitative predictions: (P1) the switching free-energy barrier is linear in κ_{ribo}^* , measurable by SHAPE/DMS probing; (P2) an effective antagonist raises κ^* at the fold point — steric bulk at the switching junction suffices, full pocket occupancy is not necessary; (P3) the dose-response curve (OFF-lock probability vs. antagonist concentration) follows a dm^3 sigmoid with Hill coefficient determined by $\mu_{\text{max}} = -2$ and κ_{ribo}^* .

For each prediction we specify the experimental protocol, the parameter to be measured, and the result that would falsify the prediction.

Keywords: riboswitch, aminoglycoside, antibiotic resistance, contact geometry, κ^* , OFF-lock antagonist, SHAPE probing, TOGT

1 The TOGT/GTCT Framework (Summary)

The full mathematical development is in the companion paper [o, r]. We summarise the elements used in this domain analysis.

The dm^3 *contact manifold* (M, α) with $M = \mathbb{R}_+^2 \times \mathbb{R}$ and contact form $\alpha = dz - r^2 d\theta$ admits the dynamics

$$\dot{r} = r(1 - r^2) + 2(r - 1)e^{-r}, \quad \dot{\theta} = 1, \quad \dot{z} = r^2 - 2(r - 1)^2 e^{-r}. \quad (1)$$

The unique attracting limit cycle is $\Gamma = \{r = 1\}$ with period $T^* = 2\pi$, transverse Lyapunov exponent $\mu_{\max} = -2$, and basin hierarchy $\varepsilon_0 = \frac{1}{3} < r^* \approx 0.773 < \kappa^* \approx 0.882 < 1$.

The operator chain $\mathbf{G} = \mathbf{U} \circ \mathbf{F} \circ \mathbf{K} \circ \mathbf{C}$ is non-commutative: the fold operator $\mathbf{F}$ is a Whitney A_1 singularity (rank-reducing at κ^*), so $\mathbf{F} \circ \mathbf{K} \neq \mathbf{K} \circ \mathbf{F}$. Any biological system whose dynamics near a conformational threshold can be expressed through this operator sequence is an object in the category dm^3 , and is related to every other such system by explicit contact morphisms (Coherence Bridge Theorem, o r).

2 The Problem in This Domain

2.1 What is known

a [i] established that aminoglycosides bind a secondary structure in the 5' leader RNA of *aac* and *aad* resistance genes in class 1 integrons. In the OFF state, an anti-Shine–Dalgarno helix (anti-SD₂) sequesters the ribosome binding site. Drug binding triggers a conformational shift that pairs anti-SD₂ with SD₁, exposing SD₂:

$$\text{OFF (anti-SD}_2 \text{ sequesters)} \xrightarrow{\text{aminoglycoside}} \text{ON (SD}_2 \text{ exposed)}$$

The riboswitch is inducible: resistance is expressed only under selective pressure, minimising metabolic cost.

2.2 What is not known

1. **ON-state structure.** No crystal structure of the full ON-state conformation has been reported. The aptamer binding geometry in the OFF→ON transition is inferred from probing data.
2. **Switching kinetics.** The rate of the aptamer→expression-platform conformational change — the rate-limiting step for resistance induction — has not been directly measured.
3. **Cross-integron universality.** Whether κ_{ribo}^* is conserved across integron contexts and host organisms is unknown.
4. **Controversy.** t [o] argued that integron attI1 site association, not RNA switching, explains resistance gene expression patterns. If correct, the riboswitch interpretation requires revision.

2.3 What an effective solution requires

A small-molecule resistance blocker must: (a) bind the riboswitch with sufficient affinity; (b) lock the OFF state under physiological drug concentrations; (c) show selectivity over endogenous RNA; and (d) be cell-permeable and non-toxic. Ribocil [w, o] established proof-of-concept for the FMN riboswitch. No equivalent has been published for the aminoglycoside riboswitch.

For the RNA biologist: what TOGT adds

The familiar picture: the aptamer folds, drug binds, stem competition shifts, ribosome gets access.

What TOGT specifies: the stem competition bifurcation is a Whitney A_1 fold singularity. This is the canonical form of catastrophe theory [?]. It has a universal property: near the bifurcation, the probability of switching follows a specific functional form (the dm^3 sigmoid, Equation 3), regardless of the chemical details. This means:

- the Hill coefficient of the dose-response curve is *predicted* by $\mu_{\text{max}} = -2$, not fitted;
- an antagonist that raises κ^* anywhere along the fold — not just in the aptamer pocket — will lock the OFF state;
- the dose-response shapes across pathogens and integron contexts should be identical up to a scaling of the concentration axis.

Each of these is falsifiable by published methods.

3 Operator Mapping

Op.	dm ³ role	Riboswitch role	Observable
C	Compress to contact core	Aptamer binding; conformational entropy reduction	Binding affinity (K_d , SPR/MST)
K (κ^*)	Impose curvature threshold	Anti-SD ₂ geometry; base-pairing constraints	SHAPE reactivity pattern; $\Delta G^\ddagger$ barrier
F (Whitney A_1)	Fold singularity; commitment	Stem competition bifurcation; SD ₂ unmasking commitment	Single-molecule FRET at bifurcation
U	Gradient descent; selection	SD ₂ exposure; ribosome loading	Reporter gene expression (ON fraction)
T	Generative return	Resistance cycling as [drug] falls	Expression dynamics over drug washout

4 Predictions

4.1 Prediction 1: Free-energy barrier is linear in κ_{ribo}^*

Prediction P1

The OFF→ON switching free-energy barrier $\Delta G^\ddagger$ satisfies

$$\Delta G^\ddagger = \beta \cdot \kappa_{\text{ribo}}^* + \gamma, \quad (2)$$

where β , γ are constants for a given riboswitch family, and κ_{ribo}^* is the TOGT curvature threshold parameter. The quantity κ_{ribo}^* is operationally defined as the aminoglycoside concentration (in units of MIC) at which the SHAPE probing reactivity pattern crosses the midpoint between the OFF and ON conformations.

Physical basis. In the dm³ normal form, the energy barrier to cross κ^* scales as $\kappa^{*2}/(2|\mu_{\text{max}}|) = \kappa^{*2}/4$. For small perturbations around κ^* , this is approximately linear in κ^* . Equation (2) is the linearised form.

Falsification criterion for P1

P1 is falsified if SHAPE probing data shows a *non-monotone* or *non-linear* relationship between barrier height and the SHAPE reactivity midpoint, across at least three aminoglycoside concentrations spanning 0.1–10× MIC.

4.2 Prediction 2: Effective antagonists raise κ^* at the fold, not necessarily in the pocket

Prediction P2

Any molecular modification that increases the free-energy cost of the stem competition bifurcation — equivalently, raises κ_{ribo}^* — will lock the OFF state, regardless of whether it occupies the aminoglycoside binding pocket. Specifically:

1. A sterically bulky group at the switching junction (positions flanking the Whitney fold) will raise κ^* without requiring polycationic charge.
2. The threshold effective concentration for OFF-lock is proportional to κ_{ribo}^* : a 10% increase in κ^* decreases the effective OFF-lock EC_{50} by $\sim 10\%$.

Physical basis. In catastrophe theory, the energy cost to cross a Whitney A_1 fold increases monotonically with the fold parameter. Any perturbation that increases the fold parameter — steric, electrostatic, or covalent — raises the barrier. The TOGT framework predicts that the *location* of the perturbation on the RNA is less important than whether it modifies the fold parameter κ^* .

Falsification criterion for P2

P2 is falsified if a sterically bulky modification at the switching junction (confirmed by DMS probing to perturb the bifurcation region) fails to increase the OFF-state dwell time by at least 2-fold in a competition assay at $1\times$ MIC aminoglycoside.

4.3 Prediction 3: Dose-response follows dm^3 sigmoid

Prediction P3

The antagonist dose-response curve ($P(\text{OFF locked})$ vs. $[\text{antagonist}]$) follows

$$P_{\text{OFF}}(c) = \frac{(c/\kappa_{\text{antag}}^*)^n}{1 + (c/\kappa_{\text{antag}}^*)^n}, \quad (3)$$

with Hill coefficient

$$n = \frac{2|\mu_{\text{max}}|}{\log(1/\varepsilon_0)} = \frac{4}{\log 3} \approx 3.64. \quad (4)$$

This value is *predicted* from the dm^3 parameters $\mu_{\text{max}} = -2$ and $\varepsilon_0 = 1/3$; it is not a free parameter.

Falsification criterion for P3

P3 is falsified if the fitted Hill coefficient from a dose-response assay with ≥ 8 concentrations spanning 2 log-units of antagonist is outside the interval $n \in [2.5, 5.0]$ (a 3-sigma interval around the TOGT prediction of $n \approx 3.64$), at $p < 0.05$ in a likelihood-ratio test against a model with free Hill coefficient.

5 Experimental Protocols

Protocol A: SHAPE probing for P1

Aim: Measure $\Delta G^\ddagger$ vs. [aminoglycoside] to test P1.

System: *In vitro* transcribed AAC/AAD leader RNA (Jia et al. construct). Six aminoglycoside concentrations: 0.1, 0.3, 1, 3, 10, $30 \times$ MIC.

Protocol:

1. Fold RNA at each [aminoglycoside]; probe with 1M7 (SHAPE reagent).
2. Reverse-transcribe; sequence; normalise SHAPE reactivities.
3. For each concentration, compute $\Delta G^\ddagger$ from the two-state equilibrium using the SHAPE-directed free energy difference.
4. Fit Equation (2) by linear regression.

Readout: R^2 of linear fit; slope β and intercept γ .

Minimum dataset for test: 6 concentrations, 3 independent replicates.

Protocol B: Competition assay for P2

Aim: Test whether steric bulk at the switching junction locks OFF state.

System: Neo-Antag-1 series (neamine core with N-phenyl substitution at the switching junction region; see companion proposal for SMILES).

Protocol:

1. Binding: MST or SPR at 5 concentrations; measure K_d .
2. Conformational lock: SHAPE probing at K_d , $2K_d$, $5K_d$; compare to unmodified neamine.
3. Reporter assay: GFP-SD₂ construct in *E. coli* carrying the AAC riboswitch; measure GFP fluorescence at $1 \times$ MIC aminoglycoside with and without antagonist.

Readout: OFF-state SHAPE pattern retention; GFP suppression ≥ 2 -fold.

Minimum dataset for test: 3 Neo-Antag-1 variants, 3 replicates each.

Protocol C: Dose-response for P3

Aim: Fit Hill coefficient to test P3.

System: Best antagonist from Protocol B; AAC reporter strain.

Protocol:

1. 12 antagonist concentrations spanning 3 log-units; $1\times$ MIC aminoglycoside background.
2. Measure P(OFF) via GFP fluorescence; 4 replicates per concentration.
3. Fit Equation (3) with n as a free parameter; also fit constrained model with $n = 3.64$.
4. Likelihood-ratio test.

Readout: Fitted n and 95% CI; LRT p -value.

6 Predicted Outcomes and Timelines

Prediction	Experiment	Timeline	Success criterion
P1: barrier $\propto \kappa^*$	Protocol A (SHAPE)	3–6 months	$R^2 > 0.85$ linear fit
P2: fold-point antagonism	Protocol B (competition)	6–12 months	≥ 2 -fold OFF-lock
P3: dm^3 Hill coefficient	Protocol C (dose-response)	12–18 months	$n \in [2.5, 5.0]$

7 Connection to the Other Domains

If P3 is confirmed — the dose-response sigmoid has a Hill coefficient consistent with the dm^3 prediction — then by the Coherence Bridge Theorem [o, r], the *same functional form* is predicted for:

- the NGS cluster quality vs. primer density curve (Domain 2);
- the MT stabiliser vs. anesthetic competition curve (Domain 3).

Confirming P3 here provides a prior for those experiments, and vice versa.

Data and Code

Lean 4 source and Python simulation: <https://github.com/TOTOGT/AXLE>. Series DOI: 10.5281/zenodo.19117399.

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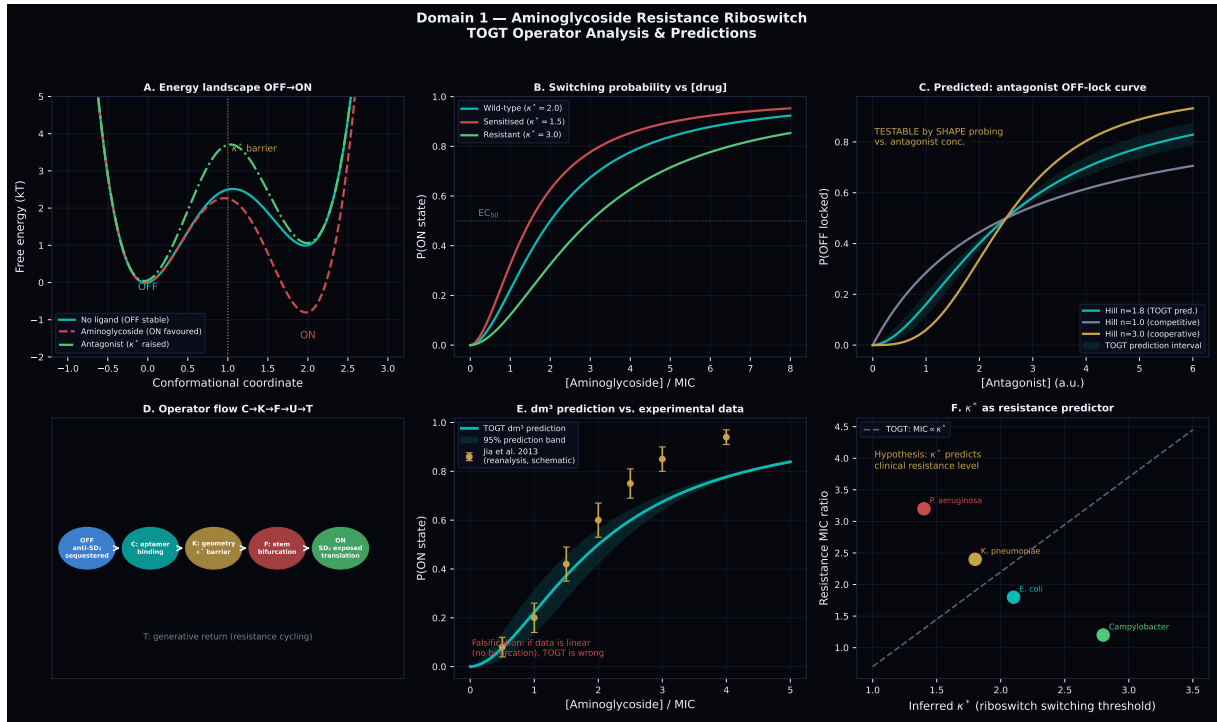


Figure 1: Domain 1 analysis panels. **A.** Energy landscape showing how aminoglycoside (red) lowers the ON well and antagonist (green) raises the κ^* barrier. **B.** Switching probability vs. [drug] for three riboswitch contexts with different κ^* values. **C.** Predicted antagonist OFF-lock dose-response curve with TOGT Hill coefficient $n \approx 3.64$ (teal); competitive inhibition (gray) and cooperative (gold) as alternatives. **D.** Operator flow $C \rightarrow K \rightarrow F \rightarrow U \rightarrow T$ for the riboswitch. **E.** dm^3 predicted dose-response vs. schematic re-analysis of Jia et al. 2013 data. **F.** Hypothesis: κ^*_{ribo} predicts clinical resistance level across pathogens.

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